

TONY WANG

US CITIZEN | 425-221-4924 | wangtzero@gmail.com | [linkedin.com/in/tony-wang-66667b242](https://www.linkedin.com/in/tony-wang-66667b242)

EDUCATION

Georgia Institute of Technology
Bachelor of Science in Computer Science
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Computer Organization and Programming, Discrete Math, Linear Algebra, Object-Oriented Programming (Java)

May 2027
Atlanta, GA

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++, SQL, Bash, Dart
Tools & Frameworks: Git, GitHub, Docker, VS Code, JIRA, Google Cloud Platform, FastAPI, React.js, Node.js
AI/Data: PyTorch, TensorFlow, Scikit-learn, Random Forest, Pandas, NumPy, Gemini API, Redis, REST APIs
Performance/Systems: Linux, Cisco UCS, NVMe, UEFI/EFI, NUMA Topology, TCP/IP, NCCL, NVLink, PCIe, Intel MLC

EXPERIENCE

Cisco
Software Engineer Intern

- Evaluated Cisco UCS server performance across CPU, memory, storage, and GPU workloads for platform validation
- Profiled CPU, memory, NUMA, and backend bottlenecks using **Intel MLC**, PCM, **AMD uProf**, perf, and SPEC CPU
- Visualized benchmark telemetry to isolate CPU, memory bandwidth, PCIe, and GPU use bottlenecks across workloads
- Validated **Kubernetes** GPU workloads with CUDA SDK pods, manifests, kubect1, nvidia-smi, and GPU Burn stress tests
- Configured server hardware across CPU swaps, **NUMA** topology, NVMe storage, and GPU validation workflows
- Analyzed **LLM inference** across tokenization, memory movement, GPU compute, KV caching, and token streaming

May 2026 – Present
San Jose, CA

Lenovo
Systems Software Engineer Intern

- Partnered with NVIDIA CUDA engineers to debug NCCL bottlenecks, improving GPU cluster efficiency by **15%** overall
- Analyzed NVLink/PCIe conflicts with HPC diagnostics, guiding targeted upgrades that removed hardware bottlenecks
- Diagnosed InfiniBand/NCCL latency with ibstat and NCCL traces, reducing distributed GPU workload runtime by **20%**
- Reduced recurring distributed workload failures by debugging Linux routing and DNS issues across GPU cluster nodes
- Automated Docker CI/CD validation workflows with Python, reducing manual testing time by **40%** overall consistently

Jun 2024 – Aug 2024
Morrisville, NC

Hong Kong University of Science and Technology
Machine Learning Engineer Intern

- Engineered Python pipelines processing **1,000+** patient records, preserving **99%** data integrity for medical AI experiments
- Optimized Random Forest regressors with cross-validation, reducing overfitting and reaching a **0.79 R²** validation score
- Refactored SQL/Bash ETL workflows for clinical datasets, reducing ingestion latency by **30%** across preprocessing runs
- Parallelized cross-validation across **256** drug combinations, accelerating ranking workflows for clinical research teams

Jun 2025 – Aug 2025
Guangzhou, China

PROJECTS

Linux Performance Profiler | C, Python, Bash, Linux, Intel MLC

- Implemented C microbenchmarks and Bash runners measuring memory, storage, and process latency across **3** workloads
- Generated Python reports for throughput, **p50/p95** latency, and run variance across **30+** benchmark trials
- Added CSV logging and threshold checks to flag **10%+** performance regressions across repeated Linux benchmark runs
- Compared CPU, memory, and NVMe bottlenecks on Linux, documenting performance tradeoffs across **3** system paths

The Dining Council | FastAPI, Redis, Docker, Streamlit, Gemini 2.5, Google Maps API

- Developed AI-powered group restaurant recommendation app with shared sessions, live chat, and **6-character** room codes
- Built FastAPI backend with **10+** REST endpoints for session management, geocoding, chat, preferences, and rankings
- Integrated Gemini 2.5 and Google Maps APIs to rank top **3** restaurants from up to **50** candidates per search
- Containerized Streamlit frontend, FastAPI backend, and Redis cache with Docker Compose, testing **10** simultaneous lobbies

Personalized Medicine AI Pipeline | PyTorch, Python, SQL

- Engineered PyTorch pipelines for treatment outcome prediction, optimizing model training on sparse clinical datasets
- Deployed low-latency inference service processing high-dimensional patient records under **50ms** for decision support
- Implemented Python imputation workflows for missing clinical data, preserving **99%** integrity for downstream modeling